

DAX AND DASHBOARD DOCUMENTATION

1. DAX Documentation

DAX (Data Analysis Expressions) was used to create calculated measures required for the analysis and to provide dynamic business insights from the Superstore dataset. The measures were designed to support the evaluation of sales performance, profitability, growth, customer and product performance, and other key business indicators.

1.1 Total Sales

The Total Sales measure calculates the overall revenue generated from all transactions in the dataset.

DAX:

Total Sales = SUM(Superstore[Sales])

This measure serves as one of the primary Key Performance Indicators (KPIs) and provides the basis for evaluating overall business performance.

1.2 Total Profit

The Total Profit measure calculates the total profit generated from all transactions.

DAX:

Total Profit = SUM(Superstore[Profit])

It is used to determine whether the revenue generated by the business is translating into actual profitability.

1.3 Total Quantity

This measure calculates the total number of units sold.

DAX:

Total Quantity = SUM(Superstore[Quantity])

The measure helps assess sales volume and understand product demand.

1.4 Profit Margin

Profit Margin measures the percentage of sales that remains as profit after costs.

DAX:

Profit Margin = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Sales}], 0)$

This measure provides a clearer understanding of profitability than total profit alone because it relates profit directly to sales revenue.

1.5 Sales Growth

Sales Growth was used to evaluate the change in sales between the current year and the previous year.

DAX:

Sales Growth =

VAR CurrentSales = [Total Sales]

VAR PreviousSales =

```
CALCULATE(  
    [Total Sales],  
    DATEADD(Superstore[Order Date], -1, YEAR)  
)
```

RETURN

$\text{DIVIDE}(\text{CurrentSales} - \text{PreviousSales}, \text{PreviousSales}, 0)$

This measure helps identify whether the business is experiencing positive or negative sales growth over time.

1.6 Profit Growth

Profit Growth measures the year-over-year change in profit.

DAX:

Profit Growth =

VAR CurrentProfit = [Total Profit]

VAR PreviousProfit =

```
CALCULATE(  
    [Total Profit],  
    DATEADD(Superstore[Order Date], -1, YEAR)  
)  
RETURN  
DIVIDE(CurrentProfit - PreviousProfit, PreviousProfit, 0)
```

It provides insight into whether improvements in sales are also resulting in improved profitability.

1.7 DAX Validation

The DAX measures were validated against the underlying dataset to ensure that the calculations returned accurate results. Measures were also tested using different dashboard filters and slicers to confirm that they responded dynamically to changes in year, category, region, segment, and other dimensions.

2. DASHBOARD DOCUMENTATION

The dashboard was developed to transform the Superstore dataset into an interactive business intelligence solution. The dashboard combines KPIs, charts, filters, and analytical visuals to provide a clear overview of sales, profitability, product performance, regional performance, and business trends.

2.1 Dashboard Objectives

The main objectives of the dashboard were to:

- * Monitor overall sales and profitability.
- * Analyse sales and profit trends over time.
- * Identify high-performing and low-performing products.

- * Evaluate category and sub-category performance.
- * Compare regional performance.
- * Identify products with high sales but relatively low profit.
- * Identify areas of potential business risk.
- * Support data-driven business decisions.

2.2 KPI Section

The KPI section provides an immediate overview of the most important business performance indicators.

The major KPIs include:

- * Total Sales – measures total revenue generated.
- * Total Profit – measures overall profitability.
- * Total Quantity – measures total units sold.
- * Profit Margin – measures profitability relative to sales.
- * Sales Growth – measures year-over-year sales performance.
- * Profit Growth – measures year-over-year profit performance.

These KPIs allow users to understand the overall health of the business before examining the detailed visuals.

2.3 Sales Trend Analysis

The sales trend visual displays how sales changed over time. It enables the identification of periods of increasing or declining sales and helps reveal seasonal patterns.

The trend analysis is particularly useful for understanding business growth and identifying periods that require further investigation.

2.4 Profitability Analysis

Profitability analysis compares sales and profit across different business dimensions. This helps identify areas where high revenue does not necessarily translate into high profitability.

The analysis is important because products or categories generating substantial sales may still have weak profit margins due to high discounts, operating costs, or other factors.

2.5 Product Performance Analysis

Product-level analysis was used to identify high-performing products as well as products requiring management attention.

Particular attention was given to products with high sales but low profit, as these products may appear successful based on revenue but may not be contributing sufficiently to the company's bottom line.

This analysis helps management review pricing, discounting, product costs, and overall product strategy.

2.6 Category and Sub-Category Analysis

The dashboard compares sales and profitability across product categories and sub-categories.

This enables the business to determine which categories contribute most significantly to revenue and profit and which categories may require corrective action.

2.7 Regional Analysis

Regional analysis evaluates business performance across different geographical regions.

It helps identify high-performing and low-performing regions and provides a basis for investigating differences in customer demand, sales volume, profitability, and market performance.

2.8 Interactive Filters and Slicers

Interactive slicers were incorporated into the dashboard to allow users to dynamically filter the analysis.

The filters can be used to examine performance by dimensions such as:

- * Year
- * Category
- * Sub-Category
- * Region
- * Segment
- * Other relevant business dimensions

When a filter is applied, the KPIs and visuals update automatically, allowing users to conduct focused analysis without modifying the underlying dataset.

2.9 Dashboard Design and Usability

The dashboard was designed with simplicity and usability in mind. Key performance indicators are positioned prominently so that users can quickly understand the overall business position.

Charts and visuals are arranged according to analytical importance, moving from overall performance to trends and then to detailed product, category, and regional analysis.

The use of consistent formatting, clear titles, appropriate chart types, and interactive filters improves readability and enables users to identify important insights efficiently.

2.10 Business Value of the Dashboard

The dashboard converts raw transactional data into actionable business information. Instead of relying solely on tables of data, decision-makers can quickly identify trends, performance gaps, profitable areas, and potential risks.

The dashboard therefore supports better decisions relating to product management, pricing, discount strategies, regional performance, sales planning, and profitability improvement.

2.11 Conclusion

The DAX calculations and interactive dashboard work together to provide a comprehensive analytical solution. DAX provides the required calculations and performance metrics, while the dashboard presents the results in an accessible and interactive format.

Overall, the solution enables management to move from simply viewing historical sales data to understanding what is happening, where it is happening, why it may be happening, and where corrective or strategic action may be required.